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Implementing a Biogeographic Ancestry Inference Service for Forensic Casework

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Abbreviations:

AF – Africa

AM – America

AISNP – Ancestry Informative Single Nucleotide Polymorphism

BGA – Biogeographic Ancestry

CFS – Centre of Forensic Sciences

EA – East Asia

EU – Europe

MPS – Massively Parallel Sequencing

OC – Oceania

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SA – South Asia

SWA – Southwest Asia

Key Words:

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Abstract

The Centre of Forensic Sciences has validated The Precision ID Ancestry Panel on the Ion S5™ Massively Parallel Sequencing instrument for use in forensic casework. The focus of this paper is the development of reporting guidelines for implementation of the biogeographic ancestry (BGA) inference service based on Admixture Prediction results produced using the Torrent Suite™ Software (Thermo Fisher Scientific). The Admixture Prediction algorithm estimates the genetic ancestry of a sample using seven root populations (Europe, East Asia, Oceania, America, Africa, South Asia, Southwest Asia). For individuals that declared a single ancestry, there was a high correlation between the declared ancestry and the ancestry predicted by the algorithm. However, some individuals with declared ancestries of Southern Europe, Southwest Asia, South Asia and the Horn of Africa had Admixture Predictions that were composed of two or more root populations at 20% or greater. For individuals with known admixed ancestry, the major component of their declaration was included in their results in all but one case. Based on these results, reporting guidelines were developed and subsequently evaluated using the Admixture Predictions of additional samples. This paper discusses the development and evaluation of these reporting guidelines, along with an implementation plan for forensic casework.

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1 Introduction

The Centre of Forensic Sciences (CFS) is a full-service forensic laboratory for investigative agencies in the province of Ontario, Canada. In our current DNA casework flow, an STR profile is generated and compared to specific individuals and/or searched against Canada's National DNA Data Bank. In the event of a non-match, the profile is of little investigative value unless a suspect comparison sample becomes available. In such cases, the ability to glean more information from the DNA, such as biogeographic ancestry (BGA), may provide an investigative aid to our clients. Although the majority of Ontarians are of European descent, according to the 2016 Census data, almost 30% of the 13.6 million residents of Ontario self-identify as members of visible minorities (defined as persons who are non-European), and an additional 3.9% self-identify as North American Aboriginal. South Asians are the largest group of visible minorities, accounting for almost 10% of Ontario's total population followed by Chinese, Black, Filipino and Arab (terminologies used here are exactly as they appear in the census). The diversity in the province's urban areas is even more pronounced, especially in Toronto (population ~2.7 million) with over 51% of the population self-identifying as visible minorities (http://www12.statcan.gc.ca/census-recensement/2016/dp-pd/prof/index.cfm?Lang=E).

Biogeographic ancestry testing has been available to police in Ontario and elsewhere, albeit on a fee for service basis through private vendors. In order to better serve its clients, the CFS acquired a Massively Parallel Sequencing (MPS) platform to initiate the validation of an ancestry panel for casework. There are two commercially available ancestry kits developed for MPS in forensics: Illumina's ForenSeq™ DNA Signature Prep Kit (targeting STRs and ancestry, identity and phenotypic informative SNPs) and Thermo Fisher's Precision ID Ancestry Panel (targeting ancestry informative SNPs only). The latter is the focus of this paper and was validated using the Ion S5™ Sequencer.

BGA inference is based on the allelic frequencies of Ancestry Informative SNPs (AISNPs) that vary among population groups and have been genotyped in a set of reference populations [1,2]. The Precision ID Ancestry

Panel (hereinafter "the Ancestry Panel") contains Kidd's 55 [2,3] and Seldin's 128 AISNPs [4,5] that have been genotyped in populations worldwide. The Torrent Suite™ Software (Thermo Fisher Scientific) can present the data using both the Population Likelihood method which calculates the likelihood of the genotype in each of the reference populations, and the Admixture Prediction method which, through

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associated algorithms, yields an ancestral composition of the unknown sample using a set of seven root populations (Thermo Fisher Scientific).

The Ancestry Panel results have been reported elsewhere for two unidentified human remains samples [6,7], twelve blind samples [8], ninety-five samples of various ancestral origin [9] and a number of population groups including Danish [10], Somali [10], Greenlanders [11], Tibeto-Burman speaking populations [12], two Japanese populations [13], Ecuadorians [14], Basques [15], North Africans [15], Turks and Iranians [16]. However, implementation guidelines for forensic casework have not yet been published. The work presented here is a subset of the validation of the MPS technology and the Ancestry Panel for casework which has been completed at the CFS. This paper outlines the steps taken to develop the reporting guidelines for the BGA inference service and its implementation strategy.

2 Materials and Methods

2.1 DNA Collection

Buccal swabs from 120 volunteers were collected for this study with informed consent. All but seven were from Ontario. They self-declared their ancestry to indicate their ancestral root populations (see Supporting Information) and provided a photograph for a visual reference. DNA was extracted using an in-house buffer (460 µl of extraction buffer:10 mM Tris pH 8, 100 mM NaCl, 1 mM EDTA pH 8, 1% SDS; 20 µl 1 M DTT, 20 µl 20 mg/µl ProK, 1 µl of 1 µg/µl cRNA) and purified on the EZ1®Advanced XL (Qiagen) using the EZ1®DNA Investigator Kit (Qiagen). Samples were quantified using the Plexor®HY System (Promega) following the manufacturer's protocol.

Additional Admixture Prediction results for 681 samples were obtained from the following external sources to evaluate the reporting guidelines developed in this study: 63 samples from Dr. David Ballard and 618 from Thermo Fisher Scientific (1000 Genomes Project through the International Genome Sample Resource; personal communications). None of the samples obtained from Thermo Fisher Scientific were used in the development of the Admixture Prediction method of the Torrent Suite™ Software Package (Thermo Fisher Scientific, personal communications).

2.2 Library Preparation

Library preparations were carried out on the Ion Chef™ Instrument (Thermo Fisher Scientific) using the Ion AmpliSeq™ Kit for Chef DL8 (Thermo Fisher Scientific) and Precision ID Ancestry Panel (Thermo

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Fisher Scientific) following the manufacturer's protocols. Eight samples (1 ng of DNA per sample) were barcoded and prepared together in each library preparation run and then pooled into one final library at completion. Quantitation of the final libraries was done using Ion Library TaqMan™ Quantitation Kit (Thermo Fisher Scientific) on the Applied Biosystems® 7500 Real-Time PCR System following the manufacturer's protocols.

2.3 Templating and Chip Loading

Both templating and chip loading were carried out on the Ion Chef™ using the Ion 520™ & 530™ Kit – Chef (Thermo Fisher Scientific) following the manufacturer's protocol. Ion 530™ Chips (Thermo Fisher Scientific) were used for all sequencing runs and 32 samples were genotyped per chip.

2.4 Sequencing and Data Analysis

The Torrent Suite™ Software v5.2.0/v5.2.2 on the Ion S5™ Sequencer (Thermo Fisher Scientific) was used to analyze the data. The runs were planned on the software following the manufacturer's protocols, selecting coverageAnalysis (v5.2.0.9) and HID_SNP_Genotyper (v4.3.2/v5.2.2) plugins to analyze the data immediately following the sequencing run using default settings. The samples genotyped by the CFS had an average coverage per SNP that ranged between 1083 and 4226x. 99.6% of the homozygous and heterozygous genotypes had Major Allele Frequency values that ranged between 95 to 100%, and 50 to 70%, respectively and 99.8% of the genotypes had the highest possible Genotype Quality value of 99.

Two methods of ancestry prediction were tested through the HID_SNP_Genotyper plugin: Population Likelihood and Admixture Prediction. The Population Likelihood method calculates a likelihood score for each of the 65 reference populations by multiplying the genotype frequencies at all loci, and then ranks the populations by their likelihood scores from the highest to the lowest (Thermo Fisher Scientific).

The Admixture Prediction method provides the ancestral composition of a sample using seven root populations: Europe (EU), Oceania (OC), East Asia (EA), Africa (AF), South Asia (SA), America (AM), and Southwest Asia (SWA). These root populations are formed through distinct groupings of 66 reference populations (one additional population compared to the Population Likelihood method) that form following a hierarchical clustering based on their allele frequencies. Fifteen of the 66 reference populations, here referred to as intermediate populations, did not cluster with any of the root populations and therefore were not used in the Admixture Prediction method (Thermo Fisher Scientific, personal communications). The

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Admixture Prediction provides the user with a percent composition of up to four root populations, in 5% increments, that best describes the genetic ancestry of a sample. A snapshot of a typical result yielded by these two methods is shown in Figure 1. The results are from a donor that declared a 100% South East Asian ancestry, specifically from the Philippines.

3 Results and Discussion

DNA samples were obtained from 120 individuals whose declared ancestries include six of the seven root populations used in the Admixture Prediction method. The donors were from various ancestral groups that are relevant to the province of Ontario. While samples from donors of Oceanic ancestry could not be obtained, Oceania is not a well-represented population in Ontario. Self-declared ancestry (exactly as provided by the donors) and the Admixture Prediction results of each donor are listed in the Supporting Information (S1-S3). It is important to note that there is inherent uncertainty associated with self-declared ancestries, which was considered when developing the implementation strategies for this service.

The Admixture Prediction and the Population Likelihood methods were both considered for casework. However due to the limited number of reference populations and the inability of the Population Likelihood method to accurately assign ancestral origins to admixed samples (results not shown), the Admixture Prediction method was chosen solely for the BGA inference service at the CFS.

3.1 Donors with a Single Self-Declared Ancestry

The Admixture Prediction results from seventy-five donors who declared that their ancestry is from one root population only are summarized in Table 1 (see Supporting Information, S1). For each root population, the percentage range of the predicted ancestry that agreed with the declaration is listed. For example, for the eleven donors that declared 100% East Asian ancestry, the East Asian component of their Admixture Prediction was between 80 and 100%.

Those that declared European or African ancestry were divided into sub populations: Northern Europe, Southern Europe and Ashkenazi Jewish for the European root population; Sub-Saharan Africa, Horn of Africa and North Africa for the African root population. Individuals that self-declared as East Asian, Northern European, and Sub-Saharan African had the highest level of concordance between declared ancestry and Admixture Prediction results, which ranged between 80 and 100%. For those that declared 100% ancestry in South Asia, the South Asian component of their Admixture Prediction ranged from 35 to

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95%; the remaining components that were 20% or greater were European or Southwest Asian or both. For those that declared 100% ancestry in Southwest Asia, the Southwest Asian component of their Admixture Prediction ranged from 30 to 90%; the remaining component that was 20% or greater was European except in one donor.

For those that declared 100% ancestry in Africa or Europe, the inconsistencies in their results can be explained by the geographic locations of the underlying reference populations that were used to define the seven root populations (Figure 2). The reference populations used to define the root population of Africa are all Sub-Saharan. This is reflected in the high percentage of African Admixture Prediction for those donors in the Sub-Saharan category, 85% or greater in all but one. We grouped those that declared Caribbean ancestry under the umbrella of Sub-Saharan Africa in consideration of the trans-Atlantic slave trade that brought millions of Africans to this region [17-19]. However, the possibility of admixture still exists in the ancestry of these individuals which may explain the 70% African prediction in one Caribbean sample. This is one example that illustrates an important limitation for investigators to be aware of: inferred BGA is not necessarily related to where one's immediate family lives presently, or where they were born and raised.

The donors that declared Ethiopian and Somalian ancestry consistently had a combination of Africa and Southwest Asia in their predicted ancestry. This is likely because the two reference populations from the Horn of Africa region included in the Ancestry Panel (Ethiopian Jewish and the Somali) are intermediate populations that did not cluster with the rest of the African populations during the hierarchical clustering, and therefore were not used to define the African root population.

The donors that declared North African ancestry had 0 to 10% predicted African ancestry combined with a high level of predicted Southwest Asian ancestry and some European ancestry. Garcia et al. [15] also observed similar results by testing two individuals from Morocco and Algeria using the Ancestry Panel. This can be attributed not only to the lack of North African reference populations used in the Ancestry Panel (Figure 2), but also to the substantial genetic separation of North Africans from the Sub-Saharan populations [20-22]. The movement of the North African populations has mainly been in an east-west direction due to the physical barrier set by the Sahara Desert and it has been proposed that most of the North Africa's genetic ancestry is derived from regions outside of Africa [22]. To provide greater clarity in our reported results and to better reflect the results seen in these donors, we have renamed the root

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population of Southwest Asia as Middle East/North Africa, and the African root population as Sub-Saharan Africa (Figure 2).

Similar geographic distinction can be made for the European donors. The underlying reference populations used to define the European root population are all Northern European (Figure 2). The Southern European populations were not included and are labeled as intermediate populations along with the two Jewish populations. Thus, it was expected that the individuals with declared Northern European ancestries had a high level of predicted European ancestry (80% or greater) while the majority of individuals declaring Southern European and Ashkenazi Jewish ancestries had a combination of Europe and Southwest Asia in their predictions. This may be explained by the fact that Southern European populations have a higher genetic diversity than their Northern counterparts and this has been attributed in part to the gene flow from North Africa and the Near East [23].

Aside from the predicted ancestry combinations that were consistently observed in donors thus far, undeclared ancestries were seen at no higher than 15% in any individual that declared a single ancestry except for one. These undeclared minor components in the Admixture Predictions may indicate a lack of strong differentiation in the allele frequencies between certain root populations.

3.2 Americas – North American Indigenous Population

It was important to test individuals that are Indigenous to Ontario (dotted circle, Figure 2) because the reference populations used to define the Americas root population are limited to Central and South America (Figure 2); however, it was challenging to collect samples from donors that are 100% Indigenous. Instead, samples were obtained from eleven donors declaring at least one North American Indigenous grandparent combined with European ancestry (see Supporting Information, S2).

Eight individuals had an Americas prediction of 25 to 80%, while the remaining three had only 10 to 15%. The remaining components of the predictions that were 20% or greater for all but two donors was European, and four individuals had undeclared predictions of Oceania, East Asia or South Asia that were also 20% or greater. The low percentage of Americas in the Admixture Prediction results for the three samples may be explained by the following: underlying populations in the Ancestry Panel that are used to define the Americas root population are all located in Central and South America (Figure 2); and the Admixture Prediction method may have difficulties with assigning accurate ancestries to these admixed

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samples (discussed further in Admixture section). It should also be noted that there is inherent admixture in at least one of the three indigenous populations of Canada. Canada recognizes three populations of Indigenous peoples, First Nations, Inuit, and Métis. By definition, the people of Métis population are of mixed ancestral backgrounds, specifically between Europeans and First Nations members (https://www.canada.ca/en/indigenous-northern-affairs.html).

3.3 Admixture

Thirty-four individuals were tested whose self-declared ancestries were composed of two or more root populations (see Supporting Information, S3). Seven of these donors declared ancestry of 50% European and 50% in one other root population (hereinafter "50/50 donors"). For six of these seven, their Admixture Prediction results accurately included the two declared root populations as the only components that were above 20%. The remaining donor (NGS52) declared 50% European and 50% South Asian ancestry and the Admixture Prediction result was 75% EU, 10% EA and 15% SA. This donor's mother who declared 100% South Asian ancestry was also tested as a part of this study (see Supporting Information, S1, NGS2). Her Admixture Prediction included a lower than expected South Asian component (60%) which helps to explain the low (15%) South Asian component in the 50/50 donor.

Twenty-seven admixed donors had complex or ambiguous declarations (including those with South/Central American declaration). For this reason, the exact concordance between the declared ancestry and the Admixture Prediction results could not be determined; their Admixture Prediction results were evaluated based upon the presence of the major component of the donors' declarations. For all but one donor (NGS48), the major component of their declaration was present in their Admixture Predictions results at 25% or greater; however, for seven individuals undeclared ancestry appeared at 20% or greater. For donor NGS48, the Admixture Prediction was 95% for Europe despite a declaration which included regions in both Europe and Southwest Asia, with the Southwest Asian components comprising a larger portion of the declaration. This may be attributable to the presence of a higher than expected European component that was commonly seen in the Admixture Prediction results of Southwest Asian donors (see Supporting Information, S1) and the uncertainty in this donor's declaration.

3.4 Reporting Guidelines

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Based on the Admixture Prediction results of the 120 samples genotyped at the CFS, reporting guidelines for the BGA inference service have been developed using the seven root populations seen in Figure 2. The results will be reported based on the number of root populations that are at or above 20%. Percentage compositions of a given sample calculated by the Admixture Prediction method will not be included in the report to investigators to avoid any possible misinterpretation of the results. Instead, one of four possible results will be provided: Single Inclusion, Single Mixed Inclusion, Double Inclusion, and Uninformative Result. We will also provide an in-house generated map highlighting the region(s) of the world that reflect the ancestry of the donor of the unknown sample as a visual guide.

3.4.1 Single Inclusion

A Single Inclusion is reported when only one root population is greater than or equal to 80% and others are 15% or less.

3.4.2 Single Mixed Inclusion

A Single Mixed Inclusion is reported when one root population is between 55% and 75% and the others are 15% or less. The reported result includes the possibility of the one root population as the sole ancestral population, or a mixed ancestry of this root population with at least one other root population. The other possible root population(s) is not specified in the report to avoid falsely inferring more meaning to the minor component(s).

3.4.3 Double Inclusion

A Double Inclusion is reported when two root populations are greater than or equal to 20%, the reported result includes the possibilities of either root population as the sole ancestral population, or a mixed ancestry combining both root populations.

3.4.4 Uninformative Results

When at least three root populations are greater than or equal to 20%, results will be reported as uninformative, as reporting an ancestry composed of three or four root populations to clients would have limited investigative value, and possibly be more misleading than helpful.

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These guidelines were applied back to the Admixture Prediction results of all 120 samples tested at the CFS and the results were as follows (see Supporting Information, S1-S3). For those individuals that declared 100% ancestry in a single root population, all reported results included the declared ancestries of the donors. However, most of the samples representing the intermediate populations such as Southern European, Horn of Africa and Ashkenazi Jewish populations had less discrimination in their results and these individuals were reported as Double Inclusions. Two samples that declared Southwest Asian and South Asian ancestry were reported as Uninformative Results.

For those donors whose declarations included North American Indigenous ancestry, seven samples were reported with the Americas root population, one was reported as uninformative (100% Indigenous declaration), and three samples did not include Americas in their results: one was reported as a Single Inclusion for Europe (at least 50% Indigenous declaration) and the remaining two were reported as Double Inclusions for Europe and South Asia, and for Europe and Oceania (at least 25% Indigenous declaration for both samples). These results highlight the need to educate our clients on the distinction between genetic and cultural ancestry of the Indigenous peoples of Ontario. As with all other root populations, investigators must not falsely exclude individuals solely based on the reported BGA and this information must be used in conjunction with other investigative information.

For six of the seven 50/50 donors, their declarations were accurately reflected in their reported BGA as Double Inclusions to include both root populations. The remaining donor (NGS52) was reported as a Single Mixed Inclusion to include the possibility of Europe as the sole ancestral population, or a mixed ancestry of Europe with at least one other root population.

For twenty-seven admixed donors whose declarations were ambiguous or complex, the major component of their declarations was included in the reported BGA for twenty-one donors. For donor NGS48 (#3, Figure 3) whose major declaration of Southwest Asia was not included in the Admixture Prediction, reported BGA was Single Inclusion for Europe, which represents the remaining European components of the declaration. The five remaining donors were reported as Uninformative Results despite having the major component of their declarations included in the Admixture Predictions above 20%. Based on their photographs, the admixed donors were unlikely to be excluded visually using the reported statements. Some of the admixed donors' photographs have been included in Figure 3 with their reporting statements to illustrate this point. It

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is important to emphasize to the investigators that all declared ancestries may not be reported for admixed donors, especially with minor ancestral contributions.

3.5 Testing of Reporting Guidelines

The reporting guidelines were also applied to the Admixture Prediction results of 681 samples genotyped by external laboratories. Concordance was determined based on the presence of the declared ancestry (stated exactly as provided) in the reported BGA (Table 2; detailed in Supporting Information, S4 and S5). All of the populations tested in this section are relevant to Ontario as per the 2016 Ontario Census data (http://www12.statcan.gc.ca/census-recensement/2016/dp-pd/prof/index.cfm?Lang=E).

3.5.1 Donors with a Single Ancestry

Six hundred and forty-eight test samples were from donors who declared that their ancestry is from one root population only and their results are summarized in Table 2. For 99.2% of samples (643 of 648), the reported BGA was concordant with the declared ancestry.

Of the five samples without concordant BGA reports, one Southwest Asian sample was reported as a Double Inclusion for Europe and South Asia. Similar results were also observed for two samples tested by Al-Asfi et al. [9]. This combination was not observed in samples processed by the CFS – all donors who declared Southwest Asian ancestry were reported with BGA that included Middle East/North Africa, except for one with Uninformative Result. Going forward, the possibility of a Southwest Asian donor should be considered when the reported BGA includes Europe and South Asia. Four remaining samples were reported as Uninformative Results. One sample was from an American donor with African ancestry. The remaining three samples were from donors of Southwest Asian or South Asian ancestries and similar results were observed with two samples tested at the CFS (see Supporting Information, S1).

3.5.2 Donors with Admixed Declarations

Twenty two samples had declarations of South/Central American ancestry and results were as follows: eight samples included Americas in their reported BGA; eight samples did not include Americas in their reported BGA, instead Europe was reported in combination with Oceania, South Asia or Middle East/North Africa; and six samples were reported as Uninformative Results (Table 2). Based on the declarations obtained from 1000 Genomes Project for these samples (see Supporting Information, S5), it was not possible to ascertain

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whether or not their genetic ancestry was fully indigenous to the Americas or an admixture, as seen with the South/Central American samples tested at the CFS (see Supporting Information, S3).

The remaining eleven samples were from 50/50 donors (see Supporting Information, S4). Seven samples reported BGAs concordant with their declarations and three were reported as Uninformative Results. The remaining sample was reported as a Single Inclusion for one of the declared root populations (the donor declared 50% Iranian and 50% Channel Islands and was reported as a Single Inclusion for Europe). It was common to see a Double Inclusion for Europe and Middle East/North Africa for those tested at the CFS who declared 100% ancestry in Southwest Asia (see Supporting Information, S1). Therefore it was not unexpected that in a donor that declared 50% ancestry in this root population, Middle East/North Africa could be omitted in their reported BGA.

3.6 Conveying Limitations and Mitigating the Risk of Misinterpretation

This service is intended for clients as an investigative aid only, in order to assist with the prioritization and collection of DNA reference samples in cases where the target crime scene profile has not matched anyone. It is important to recognize the nature of this new type of forensic DNA analysis, and the associated limitations, are very different to what clients have become accustomed to with standard STR testing. Accordingly, several measures will be put in place to ensure that results and their potential value are clearly communicated and properly understood.

Except for cases of unidentified human remains, the BGA service will only be offered where the DNA sample in question has already yielded a STR profile that has been uploaded to the Crime Scene Index of Canada's National DNA Data Bank, with no hits recorded, and where investigators otherwise have no clear suspects. From a technical standpoint, extracted DNA must remain in sufficient quantity to perform the test (at least 1ng), must be from a single-source or very strong major source of DNA, and must be of good quality as assessed by the STR profile.

In order to initiate the service, investigators will be required to submit a written request using a template provided by the laboratory. This request includes an acknowledgement of the purpose and limitations inherent to BGA testing. The limitations will be communicated to the investigators during mandatory client training provided by the CFS upon provision of test reports. Key topics of this training will include the following: the distinction between genetic ancestry and where one was born and raised or lives presently;

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the difference between genetic and cultural ancestry; the variability in appearance of individuals within root populations; the caution with which any donor's declarations must be considered, and for those with admixed ancestry, that not all components of their ancestry may appear in the report. Emphasized within this training will be that a person should not be excluded from further consideration solely on the basis of a BGA result derived from the crime scene sample, and that the only way to do so definitively is to obtain a sample from the person and compare the STR profiles.

These limitations will also be stressed in the client report, which will be disseminated and explained by forensic scientists in person with investigators. Reports will include inferred BGA of the donor of the unknown STR profile and an in-house generated map highlighting the corresponding root population(s) as indicated in Figure 2.

4 Concluding Remarks

The results of this study demonstrate the robustness of the Ancestry Panel on the Ion S5™ Sequencer for use in forensic casework. However we would suggest the following improvements to the current Ancestry Panel package as offered by the manufacturer: to allow manual elimination of genotypes with low quality scores; to allow the user to run the Admixture Prediction method as a stand-alone function by uploading a new set of genotypes; and to add new reference population data to address regions such as North Africa and North America Indigenous.

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samples that were used to test the reporting guidelines; Dr. Kenneth Kidd (Yale University School of Medicine) for his insight and support in using his population databases, ALlele FREquency Database (ALFRED; https://alfred.med.yale.edu/) and Forensic Resource/Reference on Genetics-knowledge base (FROG-kb; https://frog.med.yale.edu/); and Roger Frappier (CFS) for the technical review of this article.

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Figure 1. Example of ancestry inference results that are produced by the HID_SNP_Genotyper plugin using the Admixture Prediction and the Population Likelihood method (Thermo Fisher Scientific). Only sixteen of the sixty-five reference populations are shown here in the Population Likelihood results. The donor of this sample declared a 100% South East Asian ancestry from the Philippines.

Admixture Prediction



Population Name	Percentage
Europe	0.0
Oceania	0.0
East Asia	90.0
Africa	0.0
South Asia	10.0
America	0.0
Southwest Asia	0.0

Population Likelihood

Population Name	Geo Region	Likelihood
Ami	EastAsia	1.13E-47
Japanese-HapMap	EastAsia	4.34E-49
Lao Loum	Asia	2.81E-49
Japanese	EastAsia	4.83E-50
Koreans	EastAsia	1.12E-50
Cambodians, Khmer	EastAsia	9.39E-51
Han-HapMap	EastAsia	3.02E-51
Taiwanese Han	EastAsia	1.28E-51
Hakka	EastAsia	5.92E-52
Atayal	EastAsia	5.90E-52
Micronesians	Oceania	1.25E-53
Hazara	Asia	1.86E-56
Malaysians	EastAsia	6.97E-57
Yakut	Siberia	5.70E-57
Samoans	Oceania	1.01E-58
Kachari	Asia	8.18E-61

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Figure 2. The approximate locations of the 51 underlying populations defining the seven root populations of the Precision ID Ancestry Panel according to the hierarchical clustering results by the manufacturer (Thermo Fisher Scientific, personal communications). Each group of populations are color-coded by their respective root population. Fifteen black pins represent the intermediate populations. The root populations Southwest Asia and Africa have been renamed as Middle East/North Africa and Sub-Saharan Africa, respectively. The dotted circle indicates the province of Ontario.



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Figure 3. Examples of Individuals with admixed ancestries are summarized. The declarations are written exactly as provided by the donors. The major components of their declarations are included in the reported BGA except for #3. Based on their photographs, the donors are unlikely to be excluded visually using the reported statements (below each photograph).

1. Declared: African Ancestry; Maternal grandfather: black, Portugal; Paternal grandmother: black, England (white)	2. Declared: Mother: Mexico (great-grand parents: Spain); Father: Mexico	3. Declared: Mother: Armenia, Palestine; Father: Palestine, Russia, Poland
		
Single Inclusion for Sub-Saharan Africa	**Double Inclusion for South Asia and America**	**Single Inclusion for Europe**
4. Declared: Mother: Philippines, Spain, (maternal grandfather: Caucasian); Father: Philippines, (paternal grandfather: African)	5. Declared: Grandparents: Panama, Britain, Jamaica, Jamaica, India, Grenada, Portugal, Grenada	6. Declared : Mother: Peru, Father = Dominican Republic (to great grandparents), Spain
		
Single Inclusion for East Asia	**Double Inclusion for Sub-Saharan Africa and South Asia**	**Double Inclusion for Europe and America**

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Table 1. Summary of non-admixed samples tested for each root population and the resulting percentage range of Ancestry Prediction that was concordant with the self-declaration of the donors

	Self-declared Ancestry	Number of Samples	Percentage Range of Ancestry Prediction Concordant with the Self-Declaration (%)
	East Asia	11	80-100
	South Asia	10	35-95
	Southwest Asia	8	30-90
Europe	Northern Europe	12	80-100
Europe	Southern Europe	10	55-95
Europe	Ashkenazi Jew	8	45-100
Africa	Sub-Saharan Africa	8	70-100
Africa	Horn of Africa	6	30-70
Africa	North Africa	2	0-10

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Table 2. The reporting guidelines developed at the CFS were applied to the Admixture Prediction results of 681 test samples. Samples are grouped into root populations based on the self-declared ancestry of the donors. The number of samples that fall into the following three categories is indicated: reported BGA is concordant with self-declared ancestry; reported BGA is not concordant with self-declared ancestry; or BGA is reported as Uninformative Results. Admixture Prediction results and reported BGA for each sample can be found in Supporting Information S4 and S5

		Reported Biogeographic Ancestry		
	Self-Declared Ancestry	Concordant with Self-Declared Ancestry	Not Concordant with Self-Declared Ancestry	Uninformative Results
	100% One Root Population			
	East Asia (*n*=181)	181	0	0
	South Asia (*n*=5)	3	0	2
	Southwest Asia (*n*=7)	5	1	1
Europe	Northern Europe (*n*=174)	174	0	0
Europe	Southern Europe (*n*=97)	97	0	0
Africa	Sub-Saharan Africa (*n*=154)	154	0	0
Africa	North Africa (*n*=1)	1	0	0
Africa	Horn of Africa (*n*=5)	5	0	0
Africa	African American (*n*=24)	23	0	1
	Potentially Admixed			
	South/Central America (*n*=22)	8	8	6
	Declared Admixture (*n*=11)	7	1	3

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